Function-generated arithmetics

Catalogue, structural classification and numerical limits of the scheme
$a \oplus b = g_{\mathrm{f}}^{-1}\big(g_{\mathrm{f}}(a) + g_{\mathrm{f}}(b)\big)$

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Abstract

This note studies the family of operations $a \oplus b = g_{\mathrm{f}}^{-1}\big(g_{\mathrm{f}}(a) + g_{\mathrm{f}}(b)\big)$, where g_{f} is a bijection. Every such operation is a transport of the structure of the additive group $(\mathbb{R}, +)$ along g_{f}. Associativity, commutativity, the neutral element, additive inverses and the distributivity of the induced multiplication therefore hold automatically. We give a catalogue of ten generators with exact closed forms and classify the induced structure as a group, a monoid or a semigroup. The classification depends on the image of g_{f}, not on the shape of the formula. For the two generators without an elementary inverse (Lambert and Kepler) we measure the numerical cost. For the synergy operator with the term $2\sqrt{sX}$ we prove that for $s > 0$ it has no neutral element and is not associative, so it is not an arithmetic in the sense of Grossman and Katz; we report the measured size of those violations. We close with the physical laws that literally have this form, and with a list of claims that the scheme does not support.

Keywords: non-Newtonian calculus; transport of structure; Kolmogorov–Nagumo mean; Lambert W function; numerical verification.

1 Introduction

Many composition laws in physics share one form. First one adds a conserved quantity. Then one reads off the measured quantity. The resistance of two resistors in parallel is not the sum of the resistances. The focal length of two lenses is not the sum of the focal lengths. The root-mean-square of two voltages is not the sum of the root-mean-square values. In each case the addition happens in a different scale.

One scheme describes all of them:

$$a \oplus b = g_{\mathrm{f}}^{-1}\big(g_{\mathrm{f}}(a) + g_{\mathrm{f}}(b)\big). \tag{1}$$

The function g_{f} is a bijection. We call it the generator. The operation $\oplus$ is addition in the scale g_{f}.

The note answers three questions.

(i) Which part of (1) is a theorem, and which part is a choice?

(ii) When is the induced structure a group, and when only a monoid or a semigroup?

(iii) What happens when a term describing the environment, the so-called synergy term, is added to the sum?

The answers are largely negative. Scheme (1) does not create a new arithmetic. It transports an existing one. The synergy term leaves the algebra of groups. Neither fact discards the scheme. They place it where it is useful: as a uniform notation and as a ready set of formulas.

Table 1: What follows from Theorem 2.1 and what is a modelling choice.

Status	Item
theorem	associativity, commutativity, 0_{g_f}, additive inverses, distributivity, power and mean formulas
choice	the function g_f, the domain D, the unit and the sign of the measured quantity
separate theorem	existence of a closed form for g_f^{-1}, numerical stability, behaviour as $x \to \partial D$

2 The scheme and the transport theorem

Definition 2.1 (function-generated arithmetic)**.** Let $D \subseteq \mathbb{R}$ and let $g_\mathrm{f}\colon D \to \mathbb{R}$ be a bijection. For $a, b \in D$ we define

$$a \oplus b = g_\mathrm{f}^{-1}\big(g_\mathrm{f}(a) + g_\mathrm{f}(b)\big), \qquad a \ominus b = g_\mathrm{f}^{-1}\big(g_\mathrm{f}(a) - g_\mathrm{f}(b)\big).$$

Multiplication by a scalar $n \in \mathbb{R}$ and division by $n \neq 0$ are

$$n \odot a = g_\mathrm{f}^{-1}\big(n\, g_\mathrm{f}(a)\big), \qquad a \oslash n = g_\mathrm{f}^{-1}\left(\frac{g_\mathrm{f}(a)}{n}\right).$$

Full multiplication and full division of two elements are

$$a \otimes b = g_\mathrm{f}^{-1}\big(g_\mathrm{f}(a)\, g_\mathrm{f}(b)\big), \qquad a \oslash_f b = g_\mathrm{f}^{-1}\left(\frac{g_\mathrm{f}(a)}{g_\mathrm{f}(b)}\right).$$

Theorem 2.1 (transport of structure)**.** *The structure $(D, \oplus)$ is an abelian group. The map g_f is an isomorphism of groups $(D, \oplus) \to (\mathbb{R}, +)$.*

Proof. g_f is a bijection, so it has an inverse. For $a, b, c \in D$: $g_\mathrm{f}\big((a \oplus b) \oplus c\big) = g_\mathrm{f}(a) + g_\mathrm{f}(b) + g_\mathrm{f}(c) = g_\mathrm{f}\big(a \oplus (b \oplus c)\big)$. Commutativity follows from the commutativity of addition. The neutral element is $0_{g_\mathrm{f}} = g_\mathrm{f}^{-1}(0)$, because $g_\mathrm{f}(a \oplus 0_{g_\mathrm{f}}) = g_\mathrm{f}(a)$. The inverse of a is $\ominus a = g_\mathrm{f}^{-1}\big(-g_\mathrm{f}(a)\big)$. □

Corollary 2.1. *The following properties hold for every generator and need no separate proof: associativity, commutativity, existence of 0_{g_f}, existence of additive inverses, distributivity of $\otimes$ over $\oplus$, associativity of $\otimes$, the power formula $a^{\otimes n} = g_\mathrm{f}^{-1}\big(g_\mathrm{f}(a)^n\big)$, and the mean formula*

$$M_{g_\mathrm{f}}(a_1, \ldots, a_n) = g_\mathrm{f}^{-1}\left(\frac{1}{n}\sum_{k=1}^{n} g_\mathrm{f}(a_k)\right).$$

Corollary 2.1 is known. The mean M_{g_f} is the quasi-arithmetic mean of Kolmogorov and Nagumo [2, 3]. The family (1) was described by Grossman and Katz [1] as non-Newtonian calculus. The same scheme is the standard transport of structure in algebra [4]. The practical consequence is this. The sentence "I discovered a new arithmetic" means, in this language, "I chose a new function g_f". The content is the choice of g_f, not the formula (1).

3 Classification: group, monoid, semigroup

Theorem 2.1 assumes that g_f maps D onto all of $\mathbb{R}$. That assumption often fails. Electrical resistance is positive. Probability lies in $(0, 1)$. Absolute temperature is non-negative. The image of the generator is then a subset of $\mathbb{R}$, and the induced structure is weaker. The type of structure depends only on the image of g_f.

Proposition 3.1. *Let $g_\mathrm{f}\colon D \to \mathbb{R}$ be a bijection onto its image $g(D)$.*

Table 2: Catalogue of generators. Each row gives the exact form of g_{f}, the exact form of g_{f}^{-1}, the domain, the image, the type of structure and the constants $0_{g_{\mathrm{f}}}$ and $1_{g_{\mathrm{f}}}$. The numbers were produced by `scripts/verify.py`.

model	$g_{\mathrm{f}}(x)$	$g_{\mathrm{f}}^{-1}(y)$	D	$g(D)$	**structure**	$0_{g_{\mathrm{f}}}$	$1_{g_{\mathrm{f}}}$
harmonic	$1/x$	$1/y$	$(0,\infty)$	$(0,\infty)$	semigroup	–	–
power p	x^p	$y^{1/p}$	$[0,\infty)$	$[0,\infty)$	monoid	0	1
cubic	x^3	$\sqrt[3]{y}$	$\mathbb{R}$	$\mathbb{R}$	group	0	1
geometric	$\ln x$	e^y	$(0,\infty)$	$\mathbb{R}$	group	1	e
logistic	$\ln \frac{x}{1-x}$	$\dfrac{1}{1+e^{-y}}$	$(0,1)$	$\mathbb{R}$	group	0.5	0.7311
softplus	$\ln(e^x-1)$	$\ln(e^y+1)$	$(0,\infty)$	$\mathbb{R}$	group	0.6931	1.3133
Lambert	$x+\ln x$	$W(e^y)$	$(0,\infty)$	$\mathbb{R}$	group	0.5671	1
relativistic	$\operatorname{artanh}\frac{x}{c}$	$c\tanh y$	$(-c,c)$	$\mathbb{R}$	group	0	$0.7616c$
Kepler	$x+k\sin x$	numerically	$\mathbb{R}$	$\mathbb{R}$	group	0	0.5533
Tsallis	$\ln(1+\varepsilon x)$	$\dfrac{e^y-1}{\varepsilon}$	$(-\frac{1}{\varepsilon},\infty)$	$\mathbb{R}$	group	0	3.4366

(a) If $g(D)=\mathbb{R}$, then $(D,\oplus)$ is a group.

(b) If $g(D)=[0,\infty)$, then $(D,\oplus)$ is a monoid: $0_{g_{\mathrm{f}}}$ exists, additive inverses do not.

(c) If $g(D)=(0,\infty)$, then $(D,\oplus)$ is a semigroup: no $0_{g_{\mathrm{f}}}$ exists in D.

Proof. Part (a) is Theorem 2.1. For (b), $0 \in g(D)$, so $0_{g_{\mathrm{f}}}$ exists. For $a \neq 0_{g_{\mathrm{f}}}$ the element $-g_{\mathrm{f}}(a)$ is negative and does not belong to $g(D)$; the additive inverse does not exist. For (c), $0 \notin g(D)$, so $g_{\mathrm{f}}^{-1}(0)$ does not exist in D; the operation has no neutral element. □

Table 2 shows that two models used in practice are weaker than usually assumed. The power arithmetic with $p=2$ describes root-mean-square values and kinetic energy. It is a monoid. It has no subtraction: $a \ominus b$ exists only for $a \geq b$. The harmonic arithmetic describes equivalent resistance and focal length. It is a semigroup. It has no zero. The sentence "add zero resistance" has no meaning in that arithmetic.

4 Models that are physical laws

In this section $\oplus$ is not a modelling choice. It is the content of the law.

Parallel resistance and focal length. For the generator $g_{\mathrm{f}}(x)=1/x$ we obtain $a \oplus b = \dfrac{ab}{a+b}$. This is the formula for the equivalent resistance of two resistors in parallel and the formula for the focal length of two thin lenses. Check: $6 \oplus 3 = 2.000000$ and $50 \oplus 200 = 40.000000$.

Root-mean-square and radiation. For $g_{\mathrm{f}}(x)=x^2$ we obtain $a \oplus b = \sqrt{a^2+b^2}$. This is the root-mean-square of two uncorrelated waveforms. Example: $230 \oplus 45 = 234.360833$ volts. For $g_{\mathrm{f}}(x)=x^4$ we obtain the Stefan–Boltzmann law in the form of a sum of fluxes: $800 \oplus 1000 = 1089.617257$ kelvin. The value 1088.6 quoted in some sources is wrong.

Decibels. For $g_{\mathrm{f}}(L)=10^{L/10}$ we obtain $L_1 \oplus L_2 = 10\log_{10}\left(10^{L_1/10}+10^{L_2/10}\right)$. Two sources of 80 dB give 83.010300 dB, not 160 dB.

Table 3: Physical laws of the form (1) and their numerical results.

law	**generator**	**meaning of $\oplus$**	**example**
parallel resistance, focal length	$1/x$	sum of conductances	$6 \oplus 3 = 2$
root-mean-square	x^2	sum of powers	$230 \oplus 45 = 234.36$
Stefan–Boltzmann law	x^4	sum of fluxes	$800 \oplus 1000 = 1089.62$
sound level	$10^{L/10}$	sum of intensities	$80 \oplus 80 = 83.01$
velocity composition	$\operatorname{artanh}(v/c)$	sum of rapidities	$0.5c \oplus 0.5c = 0.8c$
independent evidence	$\ln \frac{p}{1-p}$	sum of logits	$0.8 \oplus 0.7 = 0.9032$

Composition of velocities. For $g_{\mathrm{f}}(v) = \operatorname{artanh}(v/c)$ we obtain

$$v_1 \oplus v_2 = \frac{v_1 + v_2}{1 + v_1 v_2 / c^2}.$$

This is Einstein's formula [5]. For $v_1 = v_2 = 0.5c$ the result is $0.8c$. For $v_1 = v_2 = 0.9c$ the result is $0.994475c$. The result never exceeds c, because tanh maps into $(-1, 1)$. The structure is known as a gyrovector space [6].

Pooling independent evidence. For $g_{\mathrm{f}}(p) = \ln \frac{p}{1-p}$ we obtain

$$p_1 \oplus p_2 = \frac{p_1 p_2}{p_1 p_2 + (1 - p_1)(1 - p_2)}.$$

This is the update of the odds for independent evidence in favour of an event. Example: $0.8 \oplus 0.7 = 0.903226$. Two concordant independent witnesses raise the confidence, and the result stays in $(0, 1)$.

5 Two generators without an elementary inverse

5.1 The Lambert generator

Let $g_{\mathrm{f}}(x) = x + \ln x$ for $x > 0$. The function is strictly increasing, so it is a bijection onto $\mathbb{R}$. Its inverse has the closed form $g_{\mathrm{f}}^{-1}(y) = W(e^y)$, where W is the Lambert function [8]. The constants are $0_{g_{\mathrm{f}}} = W(1) = \Omega = 0.5671432904097838$ and $1_{g_{\mathrm{f}}} = 1$.

The model has two regimes. For $a, b \ll 1$ the logarithm dominates, so

$$a \oplus b = ab\,(1 + a + b) + O\big((ab)^2\big).$$

Measurement: for $a = b = 0.01$ the relative deviation from the product is 0.0201, that is $a + b$. For $a, b \gg 1$ the identity term dominates, so $a \oplus b \to a + b$. Measurement: the relative deviation from the sum falls below 1% for $x > 25.1$ and below 0.1% for $x > 251$ (Figure 1).

The inverse g_{f}^{-1} needs care. The closed form $W(e^y)$ fails numerically. The expression e^y exceeds the double-precision range for $y > 709.78$, and W is then out of reach. The correct solution does not form e^y. It solves $x + \ln x = y$ directly, in the logarithmic domain, by Newton iteration started at $y - \ln y$. Table 4 compares the two ways.

Two conclusions. First, the asymptotic expansion has an error of $2.2 \cdot 10^{-4}$ at $y = 10$ and $1.1 \cdot 10^{-7}$ at $y = 500$. It therefore does not give machine accuracy, contrary to a claim often repeated. Second, the overflow is not a defect of W. It is a consequence of evaluating e^y where it suffices to solve the original equation.

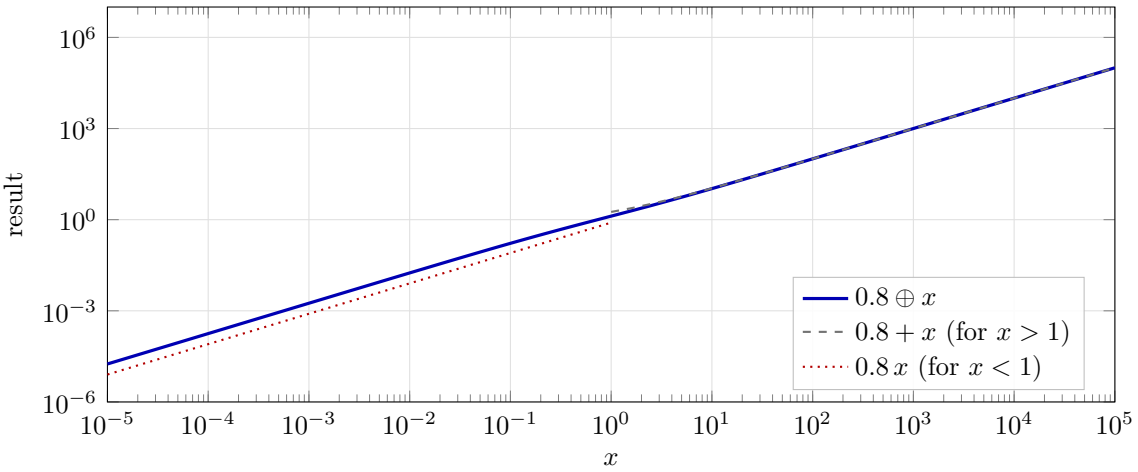


Figure 1: The Lambert arithmetic with $a = 0.8$. For $x \ll 1$ the operation coincides with multiplication. For $x \gg 1$ it coincides with addition. The transition is smooth and has no threshold. Data from `paper/data/lambert_phase.dat`.

Table 4: Computing $g_{\mathrm{f}}^{-1}(y)$ for the Lambert generator. The column "closed form" is $W(e^{y})$ in IEEE-754 double precision. The column "asymptotic" is $y - \ln y + \ln y / y$, the approximation used in the source code.

y	closed form $W(e^{y})$	solve in ln domain	asymptotic error
10	7.9294200950	7.9294200950	$2.2 \cdot 10^{-4}$
100	95.4414866456	95.4414866456	$6.3 \cdot 10^{-6}$
500	493.7978737290	493.7978737290	$1.1 \cdot 10^{-7}$
700	693.4583088790	693.4583088790	$4.4 \cdot 10^{-8}$
710	overflow	703.4440117120	$4.2 \cdot 10^{-8}$
10^{4}	overflow	9990.7905809943	$3.3 \cdot 10^{-11}$
10^{6}	overflow	999986.1845032576	$1.2 \cdot 10^{-16}$

5.2 The Kepler generator

Let $g_{\mathrm{f}}(x) = x + k \sin x$ with $k \in (0, 1)$. The derivative $g_{\mathrm{f}}'(x) = 1 + k \cos x$ is positive, so g_{f} is a bijection $\mathbb{R} \to \mathbb{R}$. The inverse has no closed form. We compute it by Newton iteration started at y; the number of iterations to accuracy 10^{-15} is between three and five. The function is known as Kepler's equation in astronomy.

The constants of this model differ from those quoted in secondary sources. $0_{g_{\mathrm{f}}} = 0$, but $1_{g_{\mathrm{f}}} = 0.553315638360866$, not 1. Check: $g_{\mathrm{f}}(1) = 1.715250337086712$, so $g_{\mathrm{f}}(1) \neq 1$. Earlier tables that give $1_{g_{\mathrm{f}}} = 1$ are wrong.

6 The synergy operator: a negative result

This section studies the modification in which the sum depends on the environment.

Definition 6.1 (synergy operator)**.** Let $g_{\mathrm{f}} \colon D \to \mathbb{R}$ be a bijection and let $s \geq 0$ be a parameter. We define

$$a \oplus_s b = g_{\mathrm{f}}^{-1}\!\left(\left(\sqrt{g_{\mathrm{f}}(a) + g_{\mathrm{f}}(b)} + \sqrt{s} \right)^{2} - s \right). \tag{2}$$

Equivalently $a \oplus_s b = g_{\mathrm{f}}^{-1}\!\left(X + 2\sqrt{sX}\right)$ with $X = g_{\mathrm{f}}(a) + g_{\mathrm{f}}(b)$ and $X \geq 0$.

Proposition 6.1. *For $s > 0$ the operation (2) is commutative, is not associative, and has no neutral element in D.*

Table 5: Measured violations for (2) with the Lambert generator, $a = 1.3$, $b = 1.7$, $c = 2.3$. Second column: relative difference between $(a \oplus_s b) \oplus_s c$ and $a \oplus_s (b \oplus_s c)$. Third column: $|a \oplus_s 0_{g_\mathrm{f}} - a|$.

s	associativity violation	neutral-element violation
0	0	0
0.01	$1.07 \cdot 10^{-2}$	0.145
0.10	$2.83 \cdot 10^{-2}$	0.478
0.50	$4.82 \cdot 10^{-2}$	1.139
1.00	$5.75 \cdot 10^{-2}$	1.673
2.00	$6.64 \cdot 10^{-2}$	2.471
5.00	$7.64 \cdot 10^{-2}$	4.156

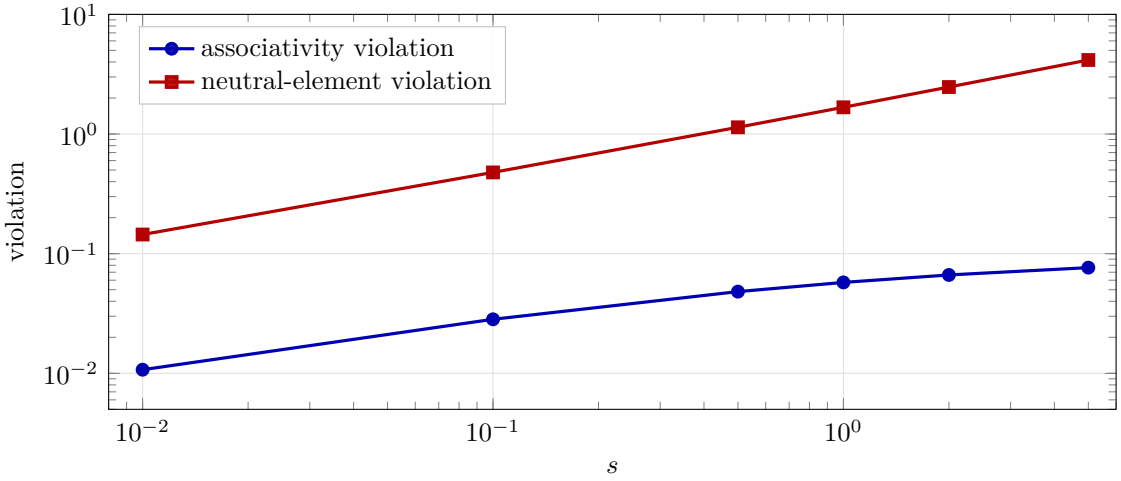


Figure 2: The violations grow like $\sqrt{s}$. The vertical axis is logarithmic. As $s \to 0$ the operator returns to the arithmetic (1).

Proof. Commutativity follows from the symmetry of X. For $a = g_\mathrm{f}^{-1}(x)$ and $b = g_\mathrm{f}^{-1}(0)$ we obtain $a \oplus_s b = g_\mathrm{f}^{-1}(x + 2\sqrt{sx}) \neq a$ for $x > 0$, so 0_{g_f} is not neutral. For associativity a numerical counterexample suffices: with the Lambert generator, $a = 1.3$, $b = 1.7$, $c = 2.3$ and $s = 0.5$ the two bracketings give 11.626886 and 12.215317; the relative difference is $4.8 \cdot 10^{-2}$. □

The conclusion is this. Operator (2) is not an arithmetic. It is not an isomorphism of $(\mathbb{R}, +)$, it has no neutral element, and it depends on the order of composition. It can be used as a one-step nonlinear update in a control loop. It cannot support the claims that it "changes the arithmetic", that it "violates conservation of energy", or that "the sum of all doses" is well defined. The last sentence matters most: without associativity there is no order-independent result for many doses.

One further observation. The term $2\sqrt{sX}$ is clipped to zero when $X < 0$. For the Lambert generator the condition $X < 0$ corresponds to $a < 0.5671$. The operator then acts as the identity. This asymmetry is an artefact of the clipping, not a physical property.

7 Claims that the scheme does not support

The list below collects statements that appear in expositions of scheme (1) and do not follow from it.

1. **Kepler's new one equals 1.** False. $1_{g_\mathrm{f}} = 0.5533$ for $k = 0.85$, and the value depends on k.
2. **Zero in the harmonic arithmetic.** It does not exist in $(0, \infty)$; $0_{g_\mathrm{f}} = \infty$ lies outside the domain.

3. **Subtraction in the power arithmetic.** It exists only for $a \geq b$; for $a < b$ the result is complex.

4. **Removal of the black-hole singularity by a logarithmic substitution.** A change of coordinates does not remove the Kretschmann invariant, which for the Schwarzschild metric equals $48G^2M^2/(c^4r^6)$ and diverges at $r = 0$ in every frame. The tortoise coordinate $r^* = r + r_s \ln|r/r_s - 1|$ is a standard tool for horizon analysis, not a model of the interior.

5. **Warp drive as a consequence of the arithmetic.** The Alcubierre metric requires negative energy density. That is an energy-cost hypothesis, not a result of the scheme.

6. **Necessity of five descriptive elements.** The number of elements is an editorial choice. No theorem forces it.

7. **Novelty of the scheme.** The scheme is from 1972 [1]. What can be new is the choice of g_{f}, the domain of application and the implementation.

8 Conclusions

Scheme (1) is a language, not a theory. Its value has three parts. First, it gives one notation for laws that look different. Second, it carries ready theorems of algebra into a new context without proof. Third, it exposes limits: monoid, semigroup and missing inverses are information about the model, not about the subject.

The synergy term is useful as a controller. It is not an arithmetic. Calling it an arithmetic leads to false statements about energy conservation and about independence from the order of composition.

Further work has three directions. First: verification of the models on measured data rather than on chosen numbers. Second: an implementation of a controller based on the saturation generator, compared with a PID controller on the same plant. Third: publication of code and data together with the text, so that every number in the tables can be reproduced by one command.

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